SFT V11.10
The Late-Time Crossing and the Sound-Horizon Bridge

A full-history reconstruction of the V11 open cosmology problem

Shaun Fosmark and ChatGPT

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Working claim. The older two-channel late-time SFT result is not discarded by the full V11 mechanism basis. It is refined. In the full-history TNG audit, the late feature becomes a smooth dominance-gradient handoff between the mass/ticking carrier and the black-hole area-growth response, with a continuous handoff band centered near $a \simeq 0.494$. A fresh CLASS thermodynamics probe gives $r_\mathrm{d} = 141.7206\,\mathrm{Mpc}$, which closes the BAO/$H_0$ bridge at $H_0 \simeq 71.40\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ for the diagnostic SFT branch. Full CMB likelihood compatibility is not claimed in this note.

Internal SFT draft. This note is a paper-style synthesis of the V11.10 late-time/BAO branch. It is not a complete CMB likelihood paper.

Contents

Abstract

Solvency Field Theory (SFT) V11 left an explicit cosmological implementation problem: whether the late-time crossing found in earlier bimodal or two-channel TNG studies survives when the full V11 mechanism basis is used. The earlier result was suggestive, but it compressed the history into a small number of effective channels and risked treating cosmic eras as switches rather than as overlapping dominance gradients.

This note reports the V11.10 reconstruction of that problem. The new pipeline treats the late universe as a smooth mixing board of channels: radiation residual, mass/Compton ticking, active astrophysical engines, black-hole area-growth response, cumulative black-hole boundary stock, and structure screening. With these channels supplied by TNG source histories and hardened by temporal/morphological null tests, the late feature survives as a smooth handoff band rather than a discrete transition. The strongest surviving families are carried by mass/ticking plus black-hole area-growth response, with additional channels stabilizing or widening the full-history band.

The resulting handoff object is then connected to the BAO/H_0 bridge through a pre-recombination CLASS calculation. A fresh patched CLASS run reports $r_\mathrm{d} = 141.7206266\,\mathrm{Mpc}$, $z_\mathrm{d} = 1059.7149$, and $\tau_\mathrm{d} = 276.0205$. Supplying this SFT sound horizon to the BAO/H_0 bridge yields $H_0 = 71.3999\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ for the diagnostic branch, a -0.0455σ pull relative to the adopted sound-horizon-free target. Opposite-sign controls remain rejected. The allowed claim is therefore a late-time/full-history TNG crossing plus CLASS-confirmed sound-horizon bridge. A full Planck/ACT/SPT likelihood analysis is deferred to a companion CMB paper.

1 Purpose and status of V11.10

The purpose of V11.10 is narrow. It does not restate the full SFT master theory, and it does not try to solve the CMB likelihood problem. It answers the open cosmology question left by the V11 master and companion chain:

> Can the late-time SFT crossing and BAO/H_0 bridge survive when the full V11 source history is used instead of the older bimodal approximation?

The answer found here is yes, with an important refinement. The physical object is not a hard transition. It is a continuous overlap band between channels whose relative dominance evolves through cosmic time.

1.1 Claim discipline

The claim level of this note is deliberately bounded.

Allowed in V11.10.

- The full-history TNG mechanism basis contains a morphology-hardened late feature near the target crossing band.
- The feature is primarily a mass/ticking plus black-hole area-growth response handoff, not a pure mass-only, BH-rate-only, or monotonic boundary-stock-only result.
- A fresh CLASS thermodynamics run gives $r_\mathrm{d} \simeq 141.7206\,\mathrm{Mpc}$ for the V12 pre-recombination branch used here.
- That sound horizon closes the BAO/H_0 bridge in the target neighborhood.

Not claimed in V11.10.

- A final DESI proof.
- A full expansion-history reconstruction with likelihood-level model selection.
- Full CMB TT/TE/EE likelihood compatibility.
- A perturbation-complete CLASS branch combining every previous CMB/source implementation.

2 From the V11 open problem to V11.10

The V11 companion introduced a corrected unified solvency ratio and a four-regime maintenance history. In that language a forced transition occurs when a dominant servicing channel reaches a critical solvency condition:

$$\Lambda_X(t) = \frac{D(t)}{C_X(t)} = \frac{H(t)I(t)}{C_X(t)}. \tag{1}$$

The earlier narrative labeled regimes as shared, independent, active, and passive. That vocabulary is useful, but it can mislead if read as a gear selector. V11.10 replaces the gear-selector picture with a mixing-board picture.

Each channel remains present after it turns on. Radiation redshifts but does not vanish. Matter/ticking becomes dominant after massive species and bound structures appear but continues into the late universe. Star formation and black-hole accretion rise and fall. Black-hole boundary stock grows monotonically. Structure screening builds as virialized systems absorb maintenance locally. The expansion history is therefore a weighted sum of fading and rising channels, not a sequence of on/off eras.

The operational picture is

$$\mathcal{F}_\mathrm{SFT}(a) = w_r(a)R(a)+w_m(a)M(a)+w_\mathrm{act}(a)A(a)+w_\mathrm{BH}(a)B_\mathrm{resp}(a)+w_\mathrm{stock}(a)B_\mathrm{stock}(a)-w_\mathrm{scr}(a)S(a), \tag{2}$$

where R is the residual radiation scaffold, M is the mass/ticking carrier, A is active astrophysical engine response, B_resp is delayed black-hole area-growth response, B_stock is cumulative black-hole boundary stock, and S is structure screening.

The old two-mix run compressed this structure. V11.10 asks whether the result survives after unpacking it.

3 Physical contrast with the older bimodal run

The older bimodal or two-channel pipeline was useful because it found a target-neighborhood crossing. But it simplified the ledger into an active channel and a passive channel. That was a good stress test, not a final physical decomposition.

3.1 Earlier effective picture

In the earlier picture, active astrophysical engines drove a changing stress term while passive structure supplied a persistent late-time depth. The result was often described as a crossing near $a \sim 0.47$.

The risk was that the late feature could be an artifact of effective compression. If active and passive channels were merely flexible basis functions, a hostile reader could ask whether the result was being tuned by representation.

3.2 Full-history V11.10 picture

The full-history audit splits the old passive idea into physically distinct components:

- direct mass/ticking carrier;
- active star-formation and black-hole accretion response;
- delayed black-hole area-growth response;
- cumulative black-hole boundary stock;
- structure screening or local absorption;
- radiation/equality/recombination scaffold.

This split changes the interpretation. The strongest families are not pure boundary stock and not pure mass. They are carried by the handoff between the mass/ticking carrier and the delayed black-hole area-growth response. Boundary stock remains present, but it is not the primary trigger of the target feature.

4 TNG source-history reconstruction

The V11.10 reconstruction uses TNG source histories as the empirical history inputs. The intent is not to fit arbitrary functions to the crossing. It is to ask whether source histories already present in the simulation carry a channel handoff in the target neighborhood.

The most important result of the full-history audit is the smooth handoff band:

Feature	Value
Smooth interval start	$a = 0.385777$
Smooth interval end	$a = 0.602387$
Smooth interval midpoint	$a = 0.494082$
Midpoint redshift	$z = 1.023955$
Width	$\Delta a = 0.216610$
Distance from target center	0.018185

The midpoint is not asserted to be the single physical crossing. It is the center of a continuous dominance-overlap interval. This distinction is central to V11.10. The physically meaningful result is the band, not merely one marker inside it.

4.1 Handoff markers

The full-history audit also produced several markers inside or near the handoff geometry:

Marker	a	Interpretation
area-response vs mass smooth midpoint	0.494082	smooth overlap interval
profile late nonlinear spine	0.499764	profile threshold marker
zero-parameter core full basis	0.501325	fixed-prior marker
area-response vs mass equal fraction	0.446800	equal-fraction marker
full V11 mechanism spine	0.505241	profile threshold marker

This marker cluster is why the correct observational comparison is a handoff band or posterior mass integral, not a single crossing point.

5 Morphology hardening and null histories

A crossing can be easy to manufacture if one allows arbitrary noisy histories. The morphology guardrail therefore asks whether the target feature is coherent. It rejects histories that land in the window only through excessive oscillation, tiny repeated crossings, or unstable phase artifacts.

The hardened morphology tests produced the following structure:

Guardrail quantity	Result
True-history hardened hits	40
Hardened null pass rate	0.045045
True candidates with null pass rate $\leq 5\%$	23/40
True candidates with null pass rate $\leq 15\%$	40/40

This does not make the result a theorem. It does make the late feature specific enough to survive hostile temporal nulls. The morphology-hardening result is one of the main differences between a suggestive run and a paper-level result.

6 Channel attribution

The attribution result is the central physics claim of the late-time branch:

> The late $a \sim 0.47 - 0.50$ feature is carried primarily by the handoff between the mass/ticking carrier and the black-hole area-growth response. Radiation residual, active engines, structure screening, and boundary stock stabilize or widen the full-history basis.

The strongest candidate families preserve this spine. A representative set is:

Family	Channels	Null pass rate
rad + mass + area response	radiation, mass/ticking, BH area response	0.015748
mass + active + area response + boundary stock + screening	mass, active, BH response, stock, screening	0.023256
mass + area response	mass/ticking, BH area response	0.023622
rad + mass + area response + screening	radiation, mass, BH response, screening	0.023622

The implication is direct: mass-only, BH-rate-only, or monotonic-stock-only stories are incomplete. The result requires the multi-channel handoff structure.

7 DESI/BAO bridge logic

The late-time branch then turns from TNG mechanism to observational bridge. The earlier BAO tests found a robust sign preference but also an amplitude/degeneracy problem: fixed-r_d BAO cannot by itself move the implied H_0 target if the standard sound horizon is imported unchanged.

That changed the question. Instead of treating $r_\mathrm{d} = 147.09\,\mathrm{Mpc}$ as a fixed input, V11.10 asks whether the SFT pre-recombination branch predicts the ruler shift demanded by the BAO/H_0 bridge.

In schematic form:

$$\text{TNG late source histories} \quad \Rightarrow \quad \varepsilon_\mathrm{SFT} < 0, \tag{3}$$

$$\text{SFT pre-recombination } H(z) \quad \Rightarrow \quad r_\mathrm{d}^\mathrm{SFT} < r_\mathrm{d}^{\Lambda\mathrm{CDM}}, \tag{4}$$

$$\text{BAO scale} + r_\mathrm{d}^\mathrm{SFT} \quad \Rightarrow \quad H_0 \text{ in the target neighborhood.} \tag{5}$$

The sign matters. Opposite-sign controls remain poor. The BAO bridge therefore sees the same direction that the TNG mechanism predicts, but the ruler shift must be supplied by the early branch rather than imported from ΛCDM.

8 CLASS sound-horizon result

A fresh CLASS-side thermodynamics probe was performed on the patched background branch. The quantity read directly from CLASS was the drag sound horizon field $r_s(z_d)$, exposed in this CLASS version as `rs_d`.

The result is:

Quantity	CLASS/SFT value
$r_{\rm d}$	141.7206266216359 Mpc
$z_{\rm d}$	1059.714863546426
$\tau_{\rm d}$	276.0204583601147

This is the bridge number. It is not a toy integral and not a regex hit from an old run. It is a fresh CLASS thermodynamics value from the patched branch used in this V11.10 chain.

9 BAO/H_0 closure

Using the CLASS-derived $r_{\rm d}$, the BAO/H_0 bridge closes in the target neighborhood.

Branch	ε	Required $r_{\rm d}$	Supplied residual	Implied H_0
Diagnostic SFT	-0.007	141.5222	+0.1984 Mpc	71.3999
Fair ΛCDM	0	142.0050	−0.2844 Mpc	71.6435
Locked SFT	-0.020	140.5611	+1.1595 Mpc	70.9150
Opposite SFT	$+0.020$	143.5091	−1.7885 Mpc	72.4023

The diagnostic SFT branch gives

$$H_0 = 71.3998804\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{6}$$

with a pull of

$$\Delta H_0 = -0.0455\sigma \tag{7}$$

relative to the adopted sound-horizon-free target.

The same bridge audit reports the opposite-sign branch as substantially worse, with a positive-sign control penalty of $\Delta\mathrm{score} \simeq 7.93$. That is the sign-specificity check. The diagnostic branch improves the raw BAO score relative to fair ΛCDM, but AIC penalizes the extra parameter, so this note does not overclaim model selection.

10 What is established

V11.10 establishes a closed late-time/BAO branch at the level appropriate for a paper-style result.

1. The full-history SFT channel basis preserves the late TNG feature.
2. The physical object is a smooth handoff band, not an on/off transition.
3. The main carrier is mass/ticking plus black-hole area-growth response.
4. Temporal and morphology guardrails reject the easiest null-history loopholes.
5. The BAO bridge prefers the SFT sign direction, while standard fixed-r_d assumptions hide the H_0 closure.
6. A fresh CLASS run supplies $r_\mathrm{d} = 141.7206\,\mathrm{Mpc}$.
7. With that ruler, the BAO/H_0 bridge lands at $H_0 \simeq 71.40\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

The result is therefore not a single numerical coincidence. It is a chain:

$$\text{TNG channels} \rightarrow \text{full-history handoff} \rightarrow \text{BAO sign} \rightarrow \text{CLASS } r_\mathrm{d} \rightarrow H_0 \text{ closure.} \tag{8}$$

11 CMB status and companion paper boundary

The CMB branch is deliberately separated from this paper. Preliminary CLASS spectra tests show the patched branch is computationally stable and retunable at the background/spectra level. However, full CMB likelihood compatibility is not claimed here.

The current state is:

CMB diagnostic quantity	Current value
Raw lensed TT RMS/base	~ 0.244
Focused retarget lensed TT RMS/base	~ 0.0985
Best retarget peak shift	0
Best morphology lensed EE full residual	~ 0.1599
Best morphology lensed TE full residual	~ 0.2630

This is enough to say the branch is not trivially broken at the CLASS-running level, and that standard retargeting reduces the TT residual substantially. It is not enough to claim Planck/ACT/SPT compatibility. The remaining residual is concentrated in damping and polarization structure, so the CMB problem deserves its own companion paper with proper likelihood machinery.

The companion CMB paper should carry:

- the V12a–V12d sound-horizon target and robustness chain;
- the V12e–V12f CLASS handoff;
- the V12h patched background implementation;
- the V12j2 direct r_d probe;
- the V12p clean patched-vs-unpatched control;
- the V12t–V12w retarget and morphology studies;
- full MCMC against Planck, ACT, and/or SPT likelihoods.

12 Predictions and near-term tests

V11.10 suggests several near-term tests.

12.1 Posterior mass in a handoff band

The observational comparison should not reduce SFT to a single crossing point. It should measure posterior mass in both a narrow target window and the full handoff band:

$$a \in [0.42, 0.52], \tag{9}$$

$$a \in [0.385777, 0.602387]. \tag{10}$$

The key markers are

$$a = 0.446800, 0.475897, 0.494082, 0.499764, 0.501325, 0.505241. \tag{11}$$

12.2 Channel-specific falsification

A hostile-reader falsification should remove or scramble specific channels:

- mass/ticking only;
- black-hole rate only;
- monotonic black-hole stock only;
- delayed area-growth response removed;
- temporal order reversed, shuffled, or phase-rolled.

The expectation is that the coherent morphology-hardened handoff should fail under these controls or lose specificity.

12.3 Sound-horizon lock

The CLASS sound horizon is now a narrow object. Rebuilds of the patched branch should reproduce

$$r_{\mathrm{d}} \simeq 141.72\,\mathrm{Mpc} \tag{12}$$

within implementation-level tolerances. If a later physical CMB profile changes r_{d} substantially, the BAO/H_0 bridge must be rerun.

13 Discussion

The main conceptual lesson of V11.10 is that SFT's late-time cosmology is not a sequence of switches. It is a changing dominance pattern among channels that never fully disappear once they are physically present.

The older bimodal run was therefore not wrong. It was compressed. It found a real target-neighborhood feature, but the full-history audit revealed the more physical carrier: a mass/ticking to black-hole area-growth response handoff, embedded in a broader channel mix.

The BAO/H_0 bridge then adds a second lesson. Late-time BAO with a fixed ΛCDM sound horizon cannot close the bridge. The pre-recombination SFT branch must supply its own ruler. CLASS does exactly that for the current branch, giving $r_\mathrm{d} = 141.7206\,\mathrm{Mpc}$ and moving the implied H_0 into the target neighborhood.

The CMB remains the correct hard test. Rather than hiding that, V11.10 separates the finished late-time/BAO result from the still-open CMB likelihood problem.

14 Conclusion

V11.10 closes the late-time crossing and sound-horizon bridge branch at the current diagnostic standard.

The full-history TNG audit shows that the late feature survives the move from the older bimodal approximation to the full V11 channel basis. The crossing is not a switch. It is a smooth dominance-gradient handoff between mass/ticking and black-hole area-growth response, stabilized by radiation residual, active engines, boundary stock, and structure screening.

The BAO bridge then requires an SFT sound horizon rather than a fixed ΛCDM ruler. A fresh CLASS thermodynamics probe returns $r_\mathrm{d} = 141.7206266\,\mathrm{Mpc}$. With that ruler, the diagnostic SFT BAO/H_0 bridge yields $H_0 = 71.3999\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, essentially on the adopted target.

The result can be stated compactly:

$$\boxed{\text{Full-history TNG handoff} + \text{CLASS } r_\mathrm{d}^\mathrm{SFT} \quad \Rightarrow \quad \text{BAO}/H_0 \text{ bridge closure.}} \tag{13}$$

The next paper should decide whether the same pre-recombination branch survives full CMB likelihood testing.

A Internal provenance summary

This paper-style draft was synthesized from the following internal result documents and checkpoints:

- SFT V11 Companion materials: forced-transition scaffold, full-history maintenance language, and V11 status discipline.
- SFT Full-History TNG Mechanism Result Note v1: full-history handoff band, morphology guardrails, channel attribution.
- V12k bridge closure audit: CLASS r_d result and BAO/H_0 closure table.
- V12u focused CMB retarget scan and V12v residual morphology audit: preliminary CMB status and companion-paper boundary.

B Minimal result table

Result	Value
Full-history smooth handoff interval	$a \in [0.385777, 0.602387]$
Handoff midpoint	$a = 0.494082$
Primary handoff carrier	mass/ticking $\leftrightarrow$ BH area-growth response
Hardened null pass rate	0.045045
CLASS drag sound horizon	$r_\mathrm{d} = 141.7206266\,\mathrm{Mpc}$
Bridge diagnostic implied H_0	$71.3999\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Opposite-sign control penalty	$\Delta\mathrm{score} \simeq 7.93$
Best preliminary retarget TT residual	~ 0.0985
CMB likelihood status	deferred